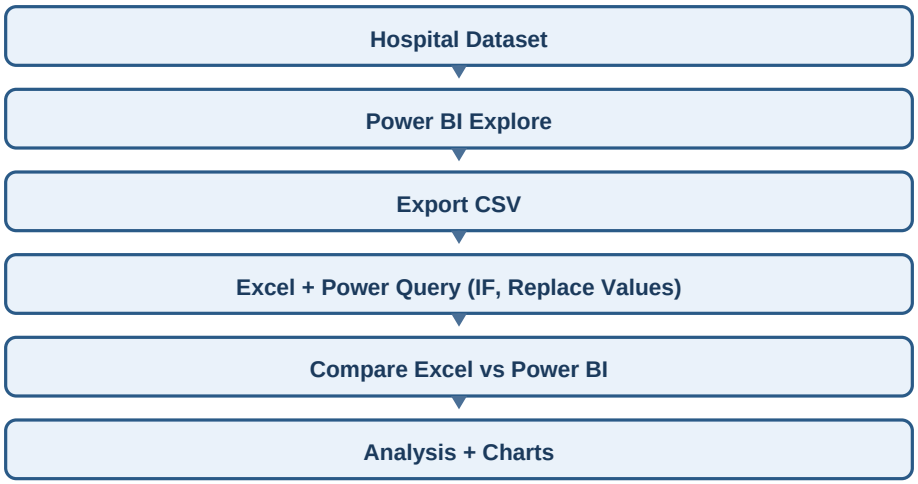
CHANDIGARH UNIVERSITY

PROJECT REPORT

EXCEL IN DATA SCIENCE

PROJECT DATA WORKFLOW



HOSPITAL EMERGENCY ROOM DATA ANALYSIS

PROJECT DOMAIN: HEALTHCARE DATA ANALYTICS

PROJECT DETAILS	INFORMATION
Prepared by	Bhumika
Programme	Bachelor of Computer Applications (BCA)
Specialisation	Data Science
Subject	Excel in Data Science
Tools used	Microsoft Power BI, Microsoft Excel, Power Query (IF statements, Replace Values), PivotTables, Formulas and Charts
Academic session	2026–2027

Submitted to Chandigarh University

SECTION 02 / INTRODUCTION TO THE PROJECT

Introduction to the Project

Using Power BI and Excel together to understand hospital emergency-room records

Healthcare organisations create large amounts of information every day. This may include patient details, admission dates, waiting times, department referrals and satisfaction scores.

Handling these records manually can be time-consuming. Data tools such as Microsoft Excel and Power BI help organise information, perform calculations and present patterns clearly.

This project studies a Hospital Emergency Room dataset. **Power BI** was used first to explore the dataset. The data was then exported as a CSV file and brought into **Excel**, where Power Query was used to clean it (IF statements and Replace Values). The waiting-time results from Excel were later **compared with the results shown in Power BI**. In total, the analysis took about two to three hours of hands-on work.

Why this project matters

- Makes patient records easier to organise and review.
- Helps examine waiting-time patterns and patient distribution.
- Demonstrates practical use of both Excel and Power BI in healthcare analytics.
- Shows how data can support informed operational decisions.

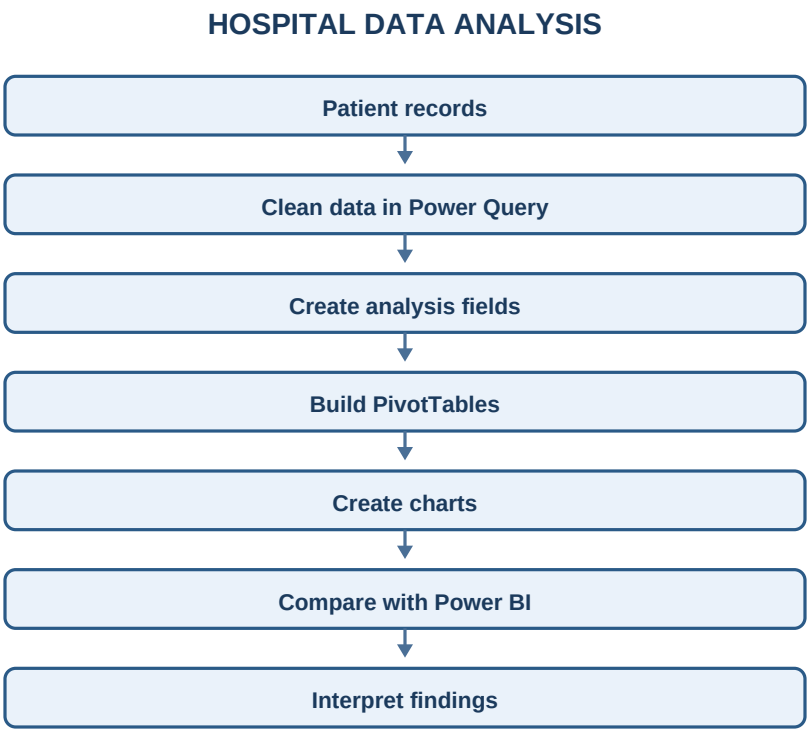
KEY IDEA The project follows a simple process: explore data, prepare it, analyse it, compare the tools and communicate the results.

SECTION 03 / OBJECTIVES OF THE PROJECT

Objectives of the Project

The main objective is to demonstrate how Excel and Power BI can be used together to analyse structured hospital data.

- Explore hospital emergency-room data in Power BI.
- Transfer data from Power BI to Excel using CSV format.
- Import and prepare data with Power Query.
- Apply IF statements and Replace Values in Power Query to clean and classify data.
- Organise patient records and review data quality.
- Analyse patient age groups and gender distribution.
- Summarise department referrals and waiting-time categories.
- Use Excel formulas for calculations and classification.
- Create PivotTables and charts to communicate results.
- Compare waiting-time results of Excel with Power BI.
- Understand how data analysis can support healthcare management.



SECTION 04 / DATASET DESCRIPTION

Dataset Description

Hospital Emergency Room Data

The dataset contains records related to patients visiting a hospital emergency department. Each row generally represents a patient visit or record, depending on the structure of the original file.

FIELD NAME	DESCRIPTION
Patient ID	Unique identifier assigned to a patient record.
Patient Gender	Gender value recorded for the patient.
Patient Age	Age of the patient.
Patient Race	Demographic category recorded in the dataset.
Patient Admission Date	Date of the hospital visit or admission.
Patient Waittime	Waiting time recorded for the patient.
Department Referral	Department to which the patient is referred.
Patient Satisfaction Score	Patient-provided satisfaction rating, if available.

DATA NOTE Column names are used as they appear in the original file. The dataset may contain extra fields or slightly different labels.

The dataset was selected because emergency-room records help illustrate patient flow, waiting-time patterns and department-wise distribution.

SECTION 05 / INTRODUCTION TO POWER BI

Introduction to Power BI

Microsoft Power BI is a business intelligence and data visualisation tool. It connects to data sources, supports data transformation and helps create reports and dashboards.

Why Power BI was used

- To open and inspect the hospital dataset.
- To understand the available columns and records.
- To explore patient-related information visually.
- To prepare the required data for export to Excel.
- To view waiting-time results visually and compare them with Excel.

Role in this project

Power BI was used first as the exploration tool, and later as a second source to cross-check the waiting-time results. Some data and visuals taken from Power BI were pasted into the Excel workbook to support the comparison.

REMEMBER Power BI supports visual exploration; Excel is used for worksheet calculations, PivotTables and charts. Using both gives a stronger analysis.

SECTION 06 / IMPORTING DATA INTO POWER BI

Importing Data into Power BI

The first practical stage was to import the Hospital Emergency Room dataset into Power BI Desktop.

1. Open Power BI Desktop.
2. Select Home → Get Data.
3. Choose the appropriate file source (for example, Text/CSV or Excel).
4. Browse to the hospital dataset and select it.
5. Preview the file and load the data.
6. Open Data view and review the available columns and records.

What was checked?

- Patient identifiers and demographic columns.
- Age and waiting-time fields.
- Admission date information.
- Department referral details.

OBSERVATION Reviewing the imported columns helps confirm that the dataset is structured and that the required fields are available.

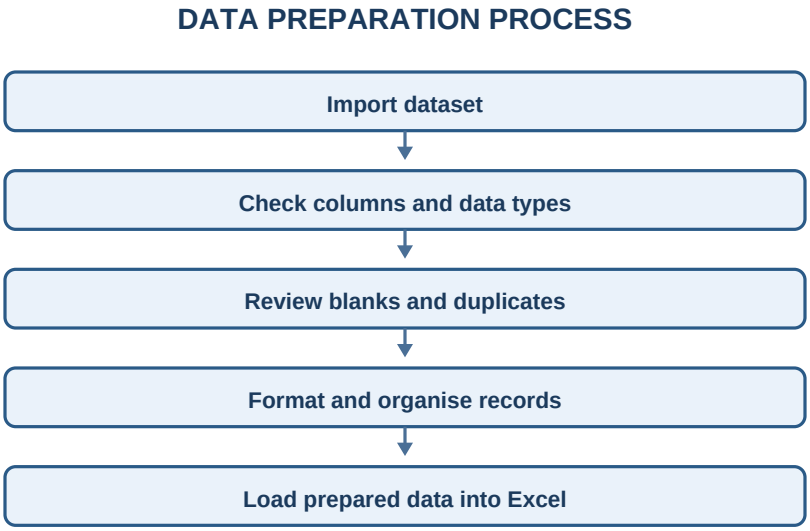
SECTION 07 / CREATING AND EXPLORING DATA IN POWER BI

Creating and Exploring Data in Power BI

After importing the file, the dataset was explored to understand its contents and identify useful fields for analysis.

Exploration checklist

- View sample patient records.
- Check column names and data types.
- Explore age and gender fields.
- Review waiting-time values.
- Examine department referral information.
- Identify fields that may be useful for charts and summaries.



Data exploration is important because it helps decide which questions can be answered using the available information.

SECTION 08 / EXPORTING DATA FROM POWER BI

Exporting Data from Power BI

After exploring the dataset, the required table data was exported so it could be analysed in Excel.

1. Open the Power BI report or table visual.
2. Select the three-dot menu (…) on the visual.
3. Choose Export data.
4. Select an available format such as CSV.
5. Select Export and save the file on the computer.

What is CSV?

CSV means Comma-Separated Values. It is a plain-text file format that stores tabular information, with values separated by commas. Excel can open and import CSV files.

Why CSV?

- Simple format for moving tabular data between applications.
- Supported by Microsoft Excel.
- Useful for structured rows and columns.

IMPORTANT Exporting a table visual exports the data represented by that visual, subject to Power BI permissions, export settings and limits.

SECTION 09 / IMPORTING POWER BI DATA INTO EXCEL

Importing Power BI Data into Excel

The exported CSV file was opened in Microsoft Excel using the Data tab.

1. Open Excel and create a blank workbook.
2. Select Data → From Text/CSV.
3. Browse to the exported CSV file.
4. Select the file and review the preview.
5. Choose Transform Data if preparation is required.
6. In Power Query Editor, select Close & Load to load the data into Excel.

Result

The exported hospital data becomes available in an Excel worksheet for further preparation, formulas, PivotTables and charts.

TIP If the CSV opens with incorrect separators or date formats, use the import preview and Power Query settings to correct them.

SECTION 10 / INTRODUCTION TO POWER QUERY

Introduction to Power Query

Power Query is a data import and transformation tool available in Excel and Power BI. It helps prepare data before loading it into a worksheet or data model.

OPTION	WHAT IT DOES
Transform Data	Opens Power Query Editor to inspect and modify the imported data.
Close & Load	Loads the prepared data into Excel using the default load settings.
Close & Load To	Lets the user choose where and how the data should be loaded.

Common Power Query tasks

- Change data types.
- Rename columns for clarity.
- Remove unnecessary columns or rows.
- Filter records.
- Replace values that are incorrect, inconsistent or missing.
- Add conditional (IF) columns to classify records.
- Review blank values and duplicates.

PROJECT ROLE Power Query acted as the preparation step between importing the CSV file and loading the dataset into Excel.

SECTION 11 / IF STATEMENTS AND REPLACE VALUES IN POWER QUERY

Using IF Statements and Replace Values in Power Query

Two important Power Query features were used to prepare the data before analysis.

1. IF statement (Conditional Column)

An IF statement checks a condition and gives one result if it is true and another if it is false. In Power Query it was added using **Add Column → Conditional Column**. It was used to classify records, for example patients who waited a short time versus a long time.

```
if [Patient Waittime] <= 30 then "On Time" else "Delayed"
```

2. Replace Values

Replace Values changes one value in a column into another value. It was used by right-clicking a column and selecting **Replace Values**. This helps make the data consistent so that PivotTables and charts group it correctly.

TASK	HOW IT HELPS
IF statement	Creates a new category column from an existing number or text column.
Replace Values	Fixes wrong, blank or inconsistently written values in a column.
Close & Load	Loads the cleaned table into Excel for further analysis.

EXAMPLE Replacing values is like correcting spellings in a list. If the same department is written in two different ways, replacing one with the other lets Excel count them as one group.

SECTION 12 / DATA CLEANING AND PREPARATION

Data Cleaning and Preparation

Data cleaning means checking a dataset for issues that could affect analysis. Examples include missing values, duplicate records, inconsistent formats and incorrect data types.

CHECK	PURPOSE
Duplicate records	Identify repeated rows or IDs that may need review.
Blank cells	Find missing information in important fields.
Replace values	Correct inconsistent or incorrect entries.
Date format	Keep dates in a consistent, recognised format.
Number format	Ensure age and waiting time are stored as numbers.
Column names	Use clear and consistent field names.
Data types	Match each field to its correct type.

Why it matters

Poor-quality data can lead to misleading counts or calculations. For example, a waiting-time field stored as text may not calculate correctly as a number.

SECTION 13 / DATA ORGANISATION AND FORMATTING IN EXCEL

Data Organisation and Formatting in Excel

After importing the dataset, the worksheet was organised to make records easier to read and analyse.

- Adjust column widths and row heights.
- Use clear, bold column headings.
- Apply suitable date and number formats.
- Use filters to view selected records.
- Keep each record in one row and each field in one column.
- Convert the range into an Excel Table where appropriate.

What is an Excel Table?

An Excel Table is a structured range of data. It provides automatic filters, consistent formatting, structured references and formula expansion when new rows are added.

GOOD PRACTICE Avoid merged cells inside the raw data table. Keep the dataset rectangular, with one header row and no completely blank rows or columns.

SECTION 14 / PATIENT AGE GROUP ANALYSIS

Patient Age Group Analysis

Age-group analysis groups individual ages into ranges. This makes it easier to compare how many patient records fall within each age category.

AGE RANGE	CATEGORY
0–20	Young patients
21–40	Adults
41–60	Middle-aged patients
61–80	Senior patients
Above 80	Elderly patients

Excel formula

If the patient's age is in cell A2, use:

```
=IF(A2<=20,"0-20",IF(A2<=40,"21-40",IF(A2<=60,"41-60",IF(A2<=80,"61-80","80+"))))
```

Fill the formula down for all patient rows. Then create a PivotTable with Age Group in Rows and Patient ID in Values (Count). A column chart can compare the number of records in each age group.

INTERPRETATION Use the completed PivotTable to identify the age group with the highest recorded number of visits. Do not assume the result before checking the data.

SECTION 15 / GENDER-WISE PATIENT ANALYSIS

Gender-wise Patient Analysis

Gender-wise analysis summarises the distribution of patient records using the gender values available in the dataset.

PivotTable setup

PIVOTTABLE AREA	FIELD
Rows	Patient Gender
Values	Patient ID — Count

Suggested visualisation

A doughnut chart or pie chart can show the proportion of records in each gender category. Include a clear title and data labels where helpful.

Why this analysis is useful

- Summarises demographic information.
- Makes category comparisons easier.
- Demonstrates how PivotTables and charts can summarise raw data.

DATA CARE Report the categories exactly as recorded. Do not infer or add categories that are not present in the source dataset.

SECTION 16 / DEPARTMENT-WISE PATIENT ANALYSIS

Department-wise Patient Analysis

Hospitals may refer emergency-room patients to different departments. Department-wise analysis counts how many records are associated with each referral category.

PivotTable setup

PIVOTTABLE AREA	FIELD
Rows	Department Referral
Values	Patient ID — Count

Suggested chart

Use a bar chart or column chart to compare department counts. Sort the values from largest to smallest if that makes the comparison clearer.

Questions to explore

- Which referral category has the most records?
- How are visits distributed across departments?
- Are some records marked as no referral or blank?

INTERPRETATION A higher count indicates more records in that category; it does not by itself prove that a department is understaffed or has poorer performance.

SECTION 17 / PATIENT WAITING TIME ANALYSIS

Patient Waiting Time Analysis

A key analysis in the project. Waiting time is the recorded time a patient waits before receiving service.

Step 1 — Create a new column

Name the new column: **Wait Status**

Step 2 — Apply the IF formula

```
=IF([@[Patient Waittime]]<=30,"On Time","Delayed")
```

RECORDED WAIT TIME	PROJECT CATEGORY
30 minutes or less	On Time
More than 30 minutes	Delayed

Step 3 — Create a PivotTable

PIVOTTABLE AREA	FIELD
Rows	Wait Status
Values	Patient ID — Count

Step 4 — Create a doughnut chart

Suggested chart title: Emergency Room Patient Waiting Time Analysis.

IMPORTANT The 30-minute cut-off is a project-defined classification for this analysis. It is not an official medical or regulatory standard.

SECTION 18 / WAITING TIME COMPARISON: EXCEL VS POWER BI

Waiting Time Comparison: Excel vs Power BI

The waiting-time analysis was done in Excel and then compared with the waiting-time view in Power BI.

What was done

1. The prepared data was kept in one Excel sheet, and the next sheet was used for the analysis.
2. The required waiting-time data was taken from the first sheet and summarised using formulas and a PivotTable in Excel.
3. The On Time and Delayed counts were calculated in Excel.
4. Waiting-time data and visuals were taken from Power BI and pasted into the Excel workbook.
5. The Excel results and the Power BI results were placed side by side and compared.

Comparison table

POINT	EXCEL RESULT	POWER BI RESULT
Total patient records	[Enter result]	[Enter result]
Average waiting time	[Enter result]	[Enter result]
On Time records	[Enter result]	[Enter result]
Delayed records	[Enter result]	[Enter result]

If both tools show the same numbers, it confirms that the data was imported, cleaned and calculated correctly. If the numbers differ, the cleaning steps, filters or data types should be checked again.

TIP Fill the table with the actual values from your PivotTable and your Power BI report, and paste a screenshot of each chart next to it.

SECTION 19 / USE OF POWER BI AND EXCEL TOGETHER

Use of Power BI and Excel Together

Both tools were used in this project, and each did a different job.

STAGE	POWER BI	EXCEL
Opening the data	Imported and explored the dataset	Imported the exported CSV file
Cleaning	Checked columns and records	Power Query: IF statements and Replace Values
Calculations	Visual summaries	Formulas, IF, COUNTIF, PivotTables
Charts	Report visuals	Column, bar, pie and doughnut charts
Final step	Waiting-time data used for comparison	Compared results and built the summary

Power BI is strong at fast visual exploration, while Excel gives full control over formulas and step-by-step calculations. Using both gave a clear check on the results.

SECTION 20 / EXCEL FUNCTIONS USED IN THE PROJECT

Excel Functions Used in the Project

Excel functions reduce manual work and help perform repeatable calculations on large datasets.

FUNCTION	PURPOSE	EXAMPLE
SUM	Adds values	=SUM(A2:A20)
AVERAGE	Calculates the mean	=AVERAGE(A2:A20)
COUNT	Counts numeric cells	=COUNT(A2:A20)
COUNTA	Counts non-empty cells	=COUNTA(A2:A20)
COUNTIF	Counts cells meeting a condition	=COUNTIF(A2:A20,">30")
IF	Returns a result based on a condition	=IF(A2>30,"Delayed","On Time")
MAX	Returns the largest value	=MAX(A2:A20)
MIN	Returns the smallest value	=MIN(A2:A20)
XLOOKUP	Looks up a value and returns a match	=XLOOKUP(E2,A2:A20,B2:B20)

NOTE Replace example ranges with the actual column or table references in your workbook. XLOOKUP availability depends on the Excel version.

SECTION 21 / PIVOTTABLES AND DATA VISUALISATION

PivotTables and Data Visualisation

A PivotTable is an Excel feature that summarises data without changing the original source records. It can calculate counts, sums, averages and other summaries.

Possible uses in this project

- Count records by gender.
- Compare patient age groups.
- Count department referrals.
- Summarise On Time and Delayed categories.
- Group admission dates by month, if the date field supports it.

CHART TYPE	POSSIBLE USE
Bar chart	Compare department counts.
Column chart	Compare age-group counts.
Doughnut chart	Show waiting-time categories.
Pie chart	Show gender distribution.
Line chart	Show monthly patient-record trends.

Why visualise data?

Charts make comparisons easier and can reveal patterns that are difficult to notice in a long table. Use clear labels, readable scales and accurate titles.

SECTION 22 / DASHBOARD AND KEY FINDINGS

Dashboard and Key Findings

A dashboard brings the most important summaries and charts together on one page.

Suggested dashboard elements

- Total patient-record count.
- Average waiting time.
- Gender distribution.
- Age-group distribution.
- Department-wise patient visits.
- On Time versus Delayed records.
- Excel vs Power BI waiting-time comparison.
- Monthly trend, if admission dates are available.

Key findings to complete from your actual workbook

QUESTION	YOUR RESULT
Which age group has the most records?	[Enter result]
Which gender category has more records?	[Enter result]
Which department has the most referrals?	[Enter result]
What is the average waiting time?	[Enter result]
How many records are classified as Delayed?	[Enter result]
Do Excel and Power BI show the same waiting-time result?	[Enter result]
Which month has the most records?	[Enter result, if analysed]

SUBMISSION TIP Replace every bracketed placeholder with a value from your own PivotTables and Power BI report. Add screenshots of your real charts where possible.

SECTION 23 / CONCLUSION AND FUTURE SCOPE

Conclusion and Future Scope

The Hospital Emergency Room Data Analysis project provided practical experience in using Microsoft Power BI and Microsoft Excel together for data analysis. The work began by exploring hospital records in Power BI. The required data was exported in CSV format and imported into Excel.

Power Query was used as the preparation step, where IF statements and Replace Values were applied before loading the data into a worksheet. The project then covered data organisation, age-group classification, gender-wise summaries, department-wise counts and waiting-time categories. Finally, the waiting-time results from Excel were compared with the data taken from Power BI.

Excel formulas, PivotTables and charts helped turn patient records into summaries that are easier to understand. The project shows how both tools can be applied to a healthcare dataset.

Future scope

- Add more records or additional hospital datasets.
- Build a more interactive Excel dashboard.
- Use advanced formulas and statistical measures.
- Explore further factors related to patient satisfaction.
- Connect Excel with Power BI for refreshed reporting.
- Develop additional Power BI visualisations.

Final learning outcome

I learned how to move data from Power BI to Excel, import and prepare a CSV file, use IF statements and Replace Values in Power Query, use formulas, create PivotTables, compare results between Excel and Power BI, and present results through charts.

PROJECT WORKFLOW Explore in Power BI → Export CSV → Import with Power Query (IF, Replace Values) → Analyse in Excel → Compare with Power BI → Present with PivotTables and charts.